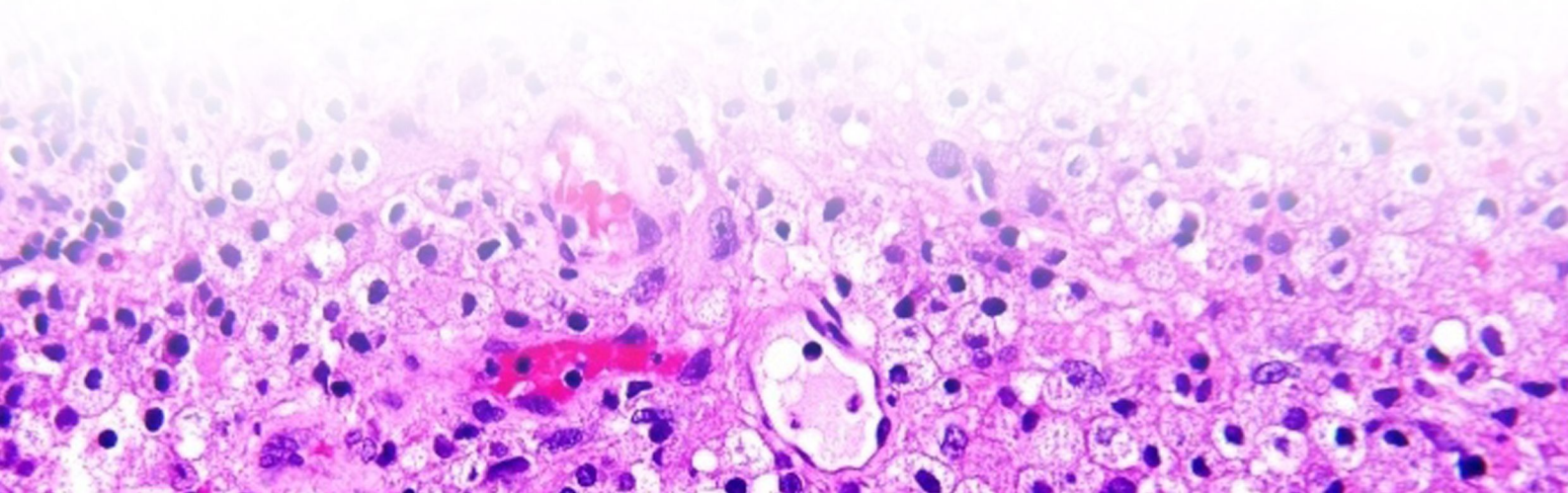
Student Notes

PATHOLOGY

1st Edition

Unit 2

Inflammation & Repair



Preface

Agam is a group of budding medicos, who are currently doing their under graduation in various Medical Colleges across Tamil Nadu and Pondicherry. The group was initiated on 18th November 2017, in the vision of uniting medicos for various social and professional causes.

We feel delighted to present you Agam Pathology notes prepared by Agam Divide and Rule 2020 Team to guide our fellow medicos to prepare for university examinations.

This is a reference work of 2017 batch medical students from various colleges. The team took effort to refer many books and make them into simple notes. We are not the authors of the following work. The images used in the documents are not copyrighted by us and is obtained from various sources.

Dear readers, we request you to use this material as a reference note, or revision note, or recall notes. Please do not learn the topics for the 1st time from this material, as this contain just the required points, for revision.

Acknowledgement

On behalf of the team, Agam would like to thank all the doctors who taught us Pathology. Agam would like to whole heartedly appreciate and thank everyone who contributed towards the making of this material. A special thanks to Vignesh M, who took the responsibility of leading the team. The following are the name list of the team who worked together, to bring out the material in good form.

HABEEB NATHIRA
HANNASHA M PRIYADARSHINI
LAVANYA E
SARAH
SHIVANGI PAL
VISHNU HARISH S
SAMYUGTHA
JENET RESHMA
VIGNESH M
MUTHAMIL SELVI E
KAUSHIK N R

INFLAMMATION AND REPAIR

ESSAY:

1. Acute inflammation
2. Mediators of acute inflammation
3. Repair and Wound healing

SHORT NOTES

1. Giant cells
2. Granulomatous inflammation
3. Leucocyte endothelial adhesion molecules
4. Role of arachidonic acid metabolites in inflammation
5. Defects in glucoside function
6. Leukotrienes
7. Chemotaxis

SHORT ANSWERS

1. Lipoxins
2. Cell derived mediators of inflammation
3. Role of Vitamin C in wound healing
4. Wound contraction
5. Wound strength
6. Acute phase reactants
7. Systemic effects of inflammation
8. Keloids
9. Exudate vs transudate

UPDATES

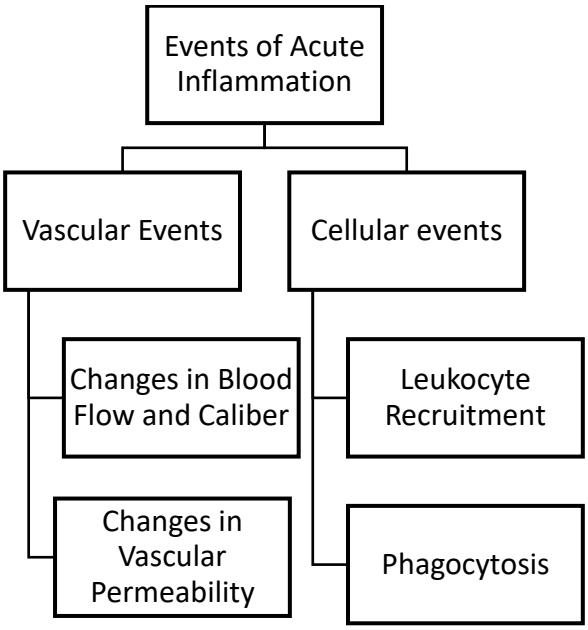
ESSAY

1. EVENTS OF ACUTE INFLAMMATION

DEFINITION

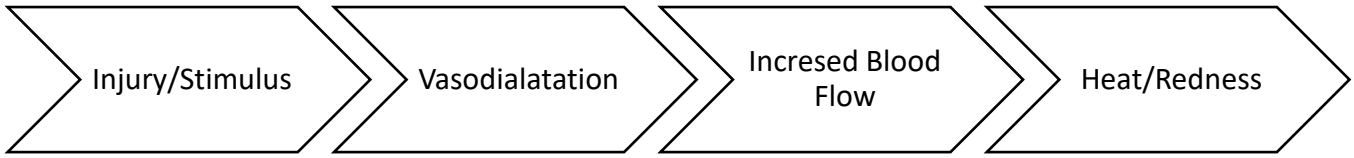
Inflammation is a response of vascularized tissues to infections and damaged tissues that brings cells and molecules of host defense from the circulation to the sites where they are needed, in order to eliminate the offending agents

EVENTS OF ACUTE INFLAMMATION

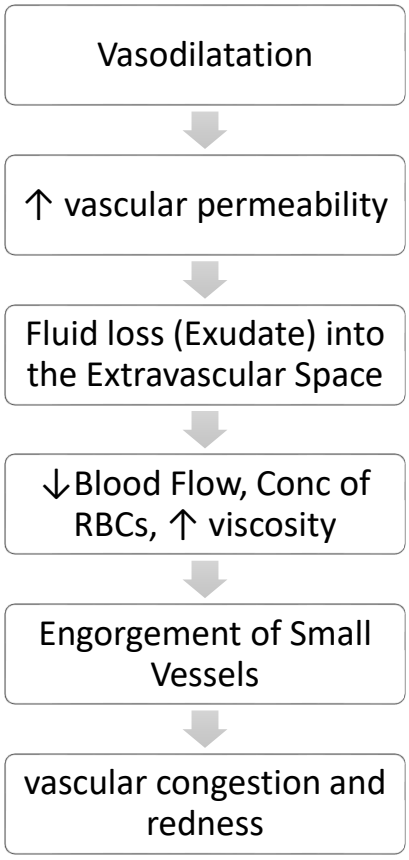


VASCULAR EVENTS OF ACUTE INFLAMMATION:

I. CHANGES IN VASCULAR FLOW AND CALIBER



II. CHANGES IN VASCULAR PERMEABILITY



MECHANISMS OF INCREASED VASCULAR PERMEABILITY:

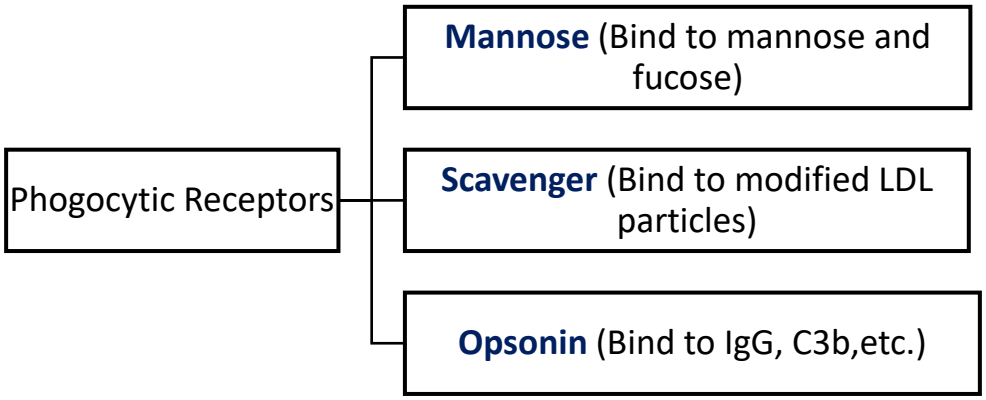
MECHANISM	MICROVASCULATURE	RESPONSE TYPE	PATHOGENESIS	EXAMPLES
1. *Endothelial cell contraction*	Venules	Immediate transient (15-30 min)	Histamine, bradykinin, others	Mild thermal injury
2. *Contraction or mild endothelial injury*	Venules, capillaries	Somewhat delayed (in 4-6 hrs) prolonged (for 24 hrs to days)	IL-1,TNF-α	Sunburns
3. *Direct endothelial cell injury*	Arterioles, venules, capillaries	Immediate prolonged (hrs to days), or delayed (2-12 hrs) prolonged (hrs to days)	Cell necrosis and detachment	Moderate to severe burns, severe bacterial infection, radiation injury
4. *Leucocyte-mediated endothelial injury*	Venules, capillaries	Delayed, prolonged	Leucocyte activation	Pulmonary venules and capillaries
5. *Neovascularisation*	All levels	Any type	Angiogenesis, VEGF	Healing, tumours

CELLULAR EVENTS OF ACUTE INFLAMMATION:

I. LEUKOCYTE RECRUITMENT

i. Margination	• Leucocyte gather along the endothelial cells.
ii. Rolling	• Mediated by SELECTINS(P,E,L) • Expression of Selectins E & P by - TNF, IL1, Selectin P- histamine & Thrombin • Ligand- Sialylated Oligosaccharides
iii. Adhesion	• Mediated by INTEGRINS • Expressoin by - TNF, IL1 • Ligand- VCAM1, ICAM1
iv. Diapedesis	• Mediate by PECAM 1
v. Chemotaxis	• Chemo attractants: Endogenous - Bacterial Products; Exogenous - Cytokines, C3a, C5a, LT-B4

II. PHAGOCYTOSIS



PROCESS: *(can be asked separately as a 4m)*

Microbes binds to one of the receptors

↓

Polymerization of actin filaments

↓

Extension of cytoplasm (pseudopods)

↓

Flows around the microbe

↓

Plasma membrane pinches off to form a vesicle (phagosome)

↓ *lysosome*

Phagolysosome

↓

Lysosomal granules

↓

Kills the microbe

KILLING MECHANISMS:

REACTIVE OXYGEN SPECIES:

$$2O_2 \rightarrow 2O_2^- + H_2 \rightarrow H_2O$$

REACTIVE NITROGEN SPECIES:

$$NO + O_2^- \rightarrow ONOO\ (Peroxy\ nitrate)$$

- Hypochlorite and Peroxy Nitrate are potent antimicrobial agents.
- Mechanism of killing is either by halogenation or oxidation.

OTHER MECHANISMS:

Contents of Lysosomal Granules

- Bactericidal permeability Increasing proteins - Activates phospholipase & degrade membrane phospholipids
- Lysozyme - Degrades bacterial coat oligosaccharides
- Major basic proteins - Cytotoxic for parasites
- Defensins - Create holes in microbial membrane

OUTCOME OF ACUTE INFLAMMATION (can be asked separately as a 4m)

All acute inflammatory reactions have one of three outcomes:

- COMPLETE RESOLUTION
- HEALING BY CONNECTIVE TISSUE REPLACEMENT
- CHRONIC INFLMMATION

1. COMPLETE RESOLUTION

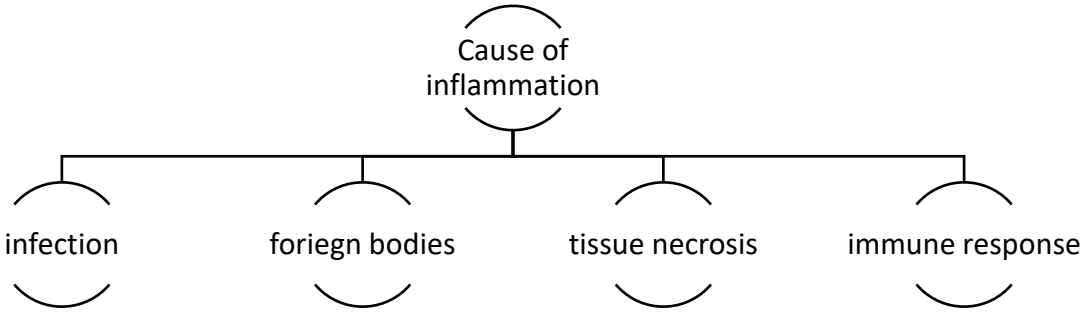
- Usual outcome when the injury is limited or short-lived.
- Clearance of injurious stimuli.
- Clearance of mediators and acute inflammatory cells.
- Replacement of injured cells.
- Ends with restoration of the site of acute inflammation to normal.
- Formation of new blood vessels and growth factor's stimulation make this transformation possible.
- Complete resolution happens only when the injured tissue is capable of regenerating.

2.HEALING BY CONNECTIVE TISSUE REPLACEMENT

- Occurs after substantial tissue destruction.
- Connective tissue grows into the area of damage or exudate.
- Scarring or fibrosis occurs.
- Scarring is the most common outcome. Fibrosis occurs when there is extensive connective tissue deposition.

3. CHRONIC INFLAMMATION

- Occurs when the acute inflammatory response can't be resolved. Ultimately results in angiogenesis, mononuclear cell infiltration and fibrosis



2. MEDIATORS OF INFLAMMATION

- The mediators of inflammation are the substances that initiate and regulate inflammatory reactions.
- The most important mediators of acute inflammation are
  - → vasoactive amines
  - → prostaglandins
  - → leukotrienes
  - → cytokines (including chemokines)
  - → products of complement activation.

PROPERTIES

- Mediators are either secreted by cells or generated from plasma proteins.
- The major cells that produce mediators of acute inflammation are **macrophages, dendritic cells** and **mast cells**. Platelets, neutrophils, endothelial cells and most epithelia can also be induced to elaborate some of the mediators.
- Plasma-derived mediators are produced mostly in the liver and are present in the circulation as inactive precursors.
- Active mediators are produced only in response to various stimuli (microbial products and substances released from necrotic cells).
- Most of the mediators are short-lived.
- One mediator can stimulate the release of other mediators.

VASOACTIVE AMINES

- Stored as preformed molecules in cells and therefore among the first mediators to be released during inflammation.

HISTAMINE

- Present in mast cells, basophils and platelets.

- Stored in mast cell granules and released by mast cell degranulation in response to a variety of stimuli:
    - Physical injury such as trauma, cold or heat
    - Binding of antibodies to mast cells, which underlies immediate hypersensitivity reactions
    - Anaphylatoxins (C3a and C5a)
    - Neuropeptides (Substance P) cytokines (IL-1, IL-8)

- Histamine causes
    - Dilation of arterioles
    - Increased vascular permeability in venules (principle mediator of immediate transient phase)
    - Contraction of some smooth muscles

SEROTONIN (5-HYDROXYTRYPTAMINE)

- Present in platelets and certain neuroendocrine cells.
- It is a neurotransmitter in the GIT.
- Causes vasoconstriction during clotting.

ARACHIDONIC ACID METABOLITES

- The lipid mediators' prostaglandins and leukotrienes are produced from arachidonic acid present in membrane phospholipids, and stimulate vascular and cellular reactions in acute inflammation.

PROSTAGLANDINS

- Produced by mast cells, macrophages, endothelial cells and many other cell types.
- Generated by actions of two cyclooxygenases, called COX-1 AND COX-2.
- Actions of some important prostaglandins:
    - $PGE_{2,}$ - Vasodilation, Increases permeability of post capillary venules,
    - PGD_2 - Chemoattractant for neutrophils
    - PGI_2 - Vasodilation, Inhibits platelet aggregation
    - TXA_2 – Vasoconstriction, Promotes platelet aggregation

LEUKOTRIENES

- Produced by leukocytes and mast cells by the action of lipoxygenase.
- Actions of some important leukotrienes:
  - LTB_4 – Potent chemotactic agent and activator of neutrophils
  - LTC_4, LTD_4, LTE_4 – Intense vasoconstriction, Increased permeability of venules and also act as bronchoconstrictor.

LIPOXINS

- Generated by the lipoxygenase pathway.
- Suppress inflammation by inhibiting recruitment of leukocytes.

CYTOKINES AND CHEMOKINES

- Cytokines are proteins produced by many cell types that mediate and regulate immune and inflammatory reactions.

CYTOKINES INVOLVED IN ACUTE INFLAMMATION ARE:

TUMORS NECROSIS FACTOR

- Produced mainly by activated macrophages, mast cells, endothelial cells and T lymphocytes.
- Stimulates expression of endothelial adhesion molecules and secretion of other cytokines.
- Involved in the systemic acute-phase response (fever, sepsis, cachexia).

INTERLEUKIN – 1

- Produced mainly by activated macrophages, endothelial cells and some epithelial cells.
- Actions are similar to that of TNF.
- It also activates tissue fibroblast.
- Has a greater role in fever.

INTERLEUKIN – 6

- Produced by macrophages and other cells.
- Induces the systemic acute-phase response.

INTERLEUKIN – 17

- Produced mainly by T lymphocytes.
- Involved in recruitment of neutrophils and monocytes.

CYTOKINES INVOLVED IN CHRONIC INFLAMMATION ARE

INTERLEUKIN – 12

- Produced by dendritic cells and macrophages.
- Stimulates the increased production of interferons.

INTERFERON-R

- Produced by T lymphocytes and NK cells.
- Activates macrophages.

INTERLEUKIN – 17

- Produced mainly by T lymphocytes.
- Involved in recruitment of neutrophils and monocytes.

CHEMOKINES

- Chemokines are a family of small proteins that act primarily as chemo-attractants for specific types of leukocytes.
- Classified into four major groups, according to the arrangement of cysteine residues:

I. **C-X-C CHEMOKINES**
   - Causes activation and chemotaxis of neutrophils.
   - Secreted by activated macrophages, endothelial cells and other cell types.
   - IL-8 is typical of this group.

II. **C-C CHEMOKINES**
   - Include MCP-1, eotaxin, MIP-1α and RANTES.
   - Attract monocytes, eosinophils, basophils and lymphocytes.

III. **C CHEMOKINES**
   - Specific for lymphocytes.

IV. **CX_3C chemokines**
   - Fractalkine is the only known member.
   - Attracts and promotes strong adhesion of monocytes and T cells.

COMPLEMENT SYSTEM:

- The complement system is a collection of soluble proteins and membrane receptors that function mainly in host defense against microbes and in pathologic inflammatory reactions.
- Functions in both innate and adaptive immunity.
- Complement proteins are present in inactive forms in the plasma.
- The critical step in complement activation is the proteolysis of the third component, C3. Cleavage of C3 can occur by:
    a) Classical pathway which is triggered by fixation of C1 to antibody that has combined with antigen.
    b) Alternative pathway which can be triggered by microbial surface molecules, in the absence of antibody.
    c) Lectin pathway in which plasma mannose-binding lectin binds to carbohydrates on microbes and directly activates C1.
- The activation of complement is tightly controlled by cell-associated and circulating regulatory proteins (C1 inhibitor, Decay Accelerating Factor, Factor H and CD-59).

MAIN FUNCTIONS OF THE COMPLEMENT SYSTEM:

a. **INFLAMMATION**

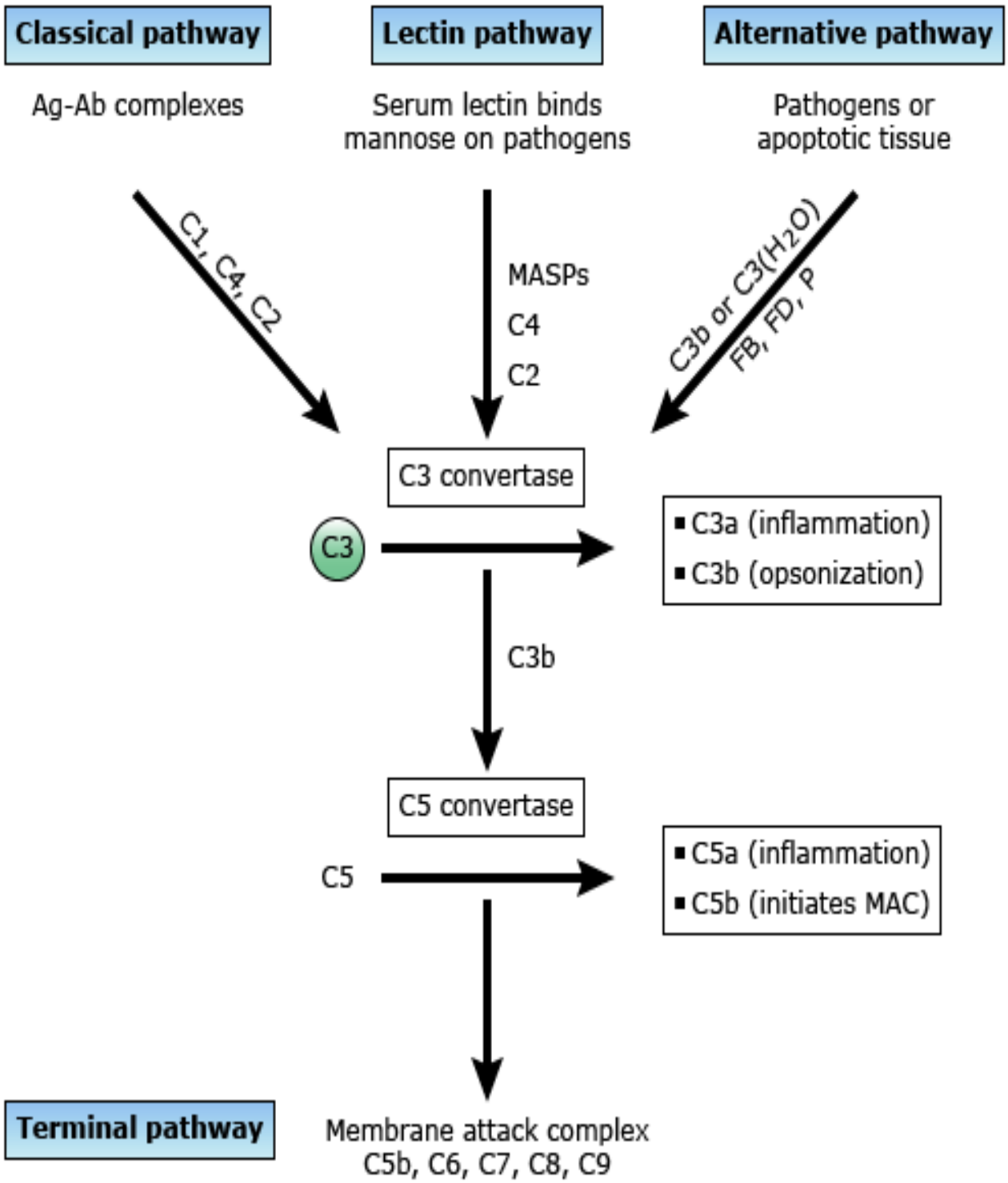
- C3a, C5a and C4a (anaphylatoxins) stimulate histamine release from mast cells resulting in increased vascular permeability and cause vasodilation.
- C5a is also chemotactic for neutrophils, monocytes, eosinophils and basophils.

b. **OPSONIZATION AND PHAGOCYTOSIS**

- C3b and iC3b, when fixed to a microbial cell wall, act as opsonins and promote phagocytosis by neutrophils and macrophages.

c. **CELL LYSIS**

- Deposition of MAC on cells makes these cells permeable to water and ions and results in death of the cells and some bacteria such as Neisseria.


Classical pathway
Lectin pathway
Alternative pathway
Ag-Ab complexes
Serum lectin binds mannose on pathogens
Pathogens or apoptotic tissue
C1, C4, C2
MASPs
C4
C2
C3b or C3(H2O)
FB, FD, P
C3 convertase
C3
C3a (inflammation)
C3b (opsonization)
C3b
C5 convertase
C5
C5a (inflammation)
C5b (initiates MAC)
Terminal pathway
Membrane attack complex
C5b, C6, C7, C8, C9

OTHER MEDIATORS OF INFLAMMATION

PLATELET-ACTIVATING FACTOR

- Phospholipid-derived mediator.
- Causes platelet aggregation, vasoconstriction and bronchoconstriction.
- At low doses, it induces vasodilation and increased vascular permeability.

KININS

- Vasoactive peptides derived from plasma proteins called kininogens, by the action of specific proteases called kallikreins.
- Bradykinin increases vascular permeability, causes contraction of smooth muscle, dilation of blood vessels and pain when injected into the skin.

NEUROPEPTIDES

- Secreted by sensory nerves and various leukocytes.
- Play a role in the initiation and regulation of inflammatory responses.
- Examples: Substance P, Neurokinin A

3. WOUND REPAIR

REGENERATION REPAIR

- Lost tissue is replaced by tissue of similar type.
- The complete restoration of lost or damaged tissue by lost/damaged tissue is replaced by fibrous tissue or scar proliferation of residual uninjured cells and replacement from stem cells.

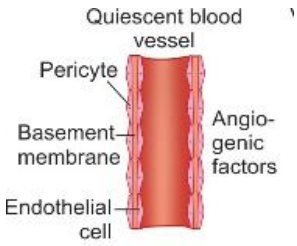
HEALING BY REPAIR

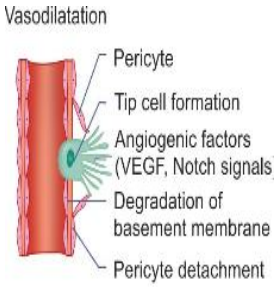
INFLAMMATION:

Inflammatory reactions limit the extent of damage.

Also promotes deposition of ECM components and stimulates angiogenesis.

ANGIOGENESIS:





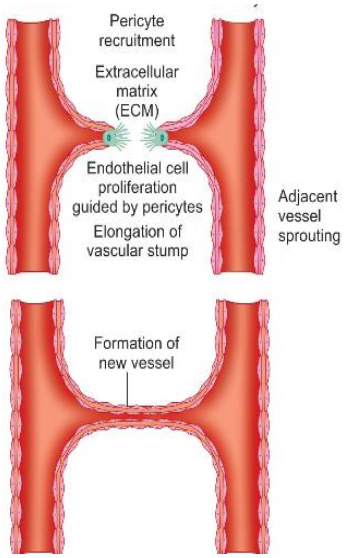
Vasodilation and increased vascular permeability.

Due to: VEGF&NO

Separation of pericytes. Breakdown of basement membrane.

Proliferation and Migration of endothelial cells

Due To: FGF



Maturation and Remodeling into capillary tubes or vessels

Due To: Angiopoietins 1&2

Peri endothelial cells recruitment.
Suppression of proliferation.
Deposition of Basement Membrane.

FORMATION OF GRANULATION TISSUE:

→ First 24 to 72 hours - Proliferation of fibroblasts and vascular endothelial cells.

→ Forms a specialized type of tissue - granulation tissue- hallmark/characteristic of tissue repair.

→ Morphology: pink, soft, granular appearance on the surface of healing wounds.

ITS CHARACTERISTIC FEATURES ARE:

- new small blood vessels (angiogenesis): new blood vessels are leaky, which allow the passage of plasma proteins and fluid into the extravascular space, which is responsible for edema.
- Proliferation of fibroblasts.

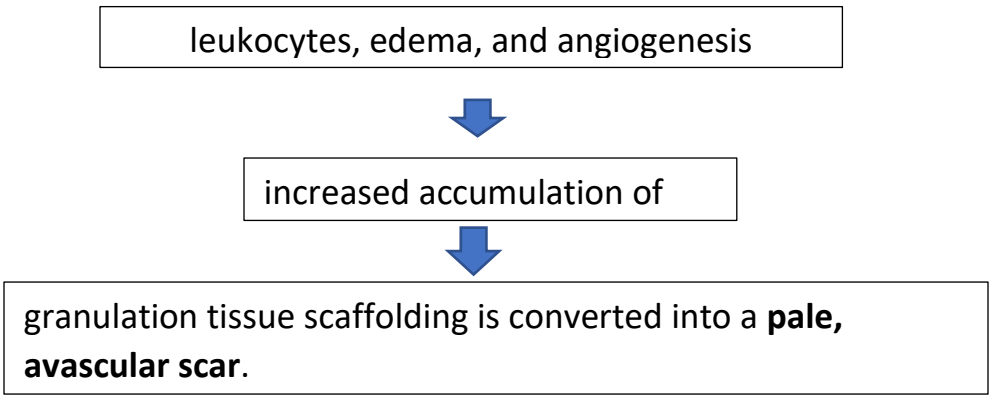
MICROSCOPICALLY, GRANULATION TISSUE CONSISTS OF:

- New small blood vessels
- Fibroblasts.
- Immune Cells Infiltration

AMOUNT OF GRANULATION TISSUE FORMED DEPENDS ON THE:

- Size of the tissue deficit created by the wound
- Intensity of inflammation.

SCAR FORMATION:



Granulation tissue is essential for repair.

COMPONENTS OF SCAR:

- spindle-shaped
- fibroblasts
- dense collagen
- fragments of elastic tissue
- other ECM components.

End of the first month - the scar consists of acellular connective tissue without inflammatory infiltrate.

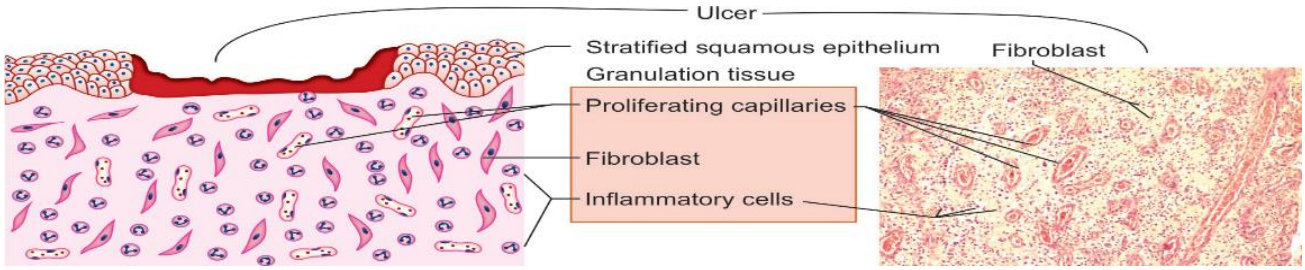
CONNECTIVE TISSUE REMODELING

- long-lasting phase of tissue repair.
- equilibrium/balance between ECM synthesis (collagen deposition) and degradation has been restored.

ROLE OF MACROPHAGES IN REPAIR

FUNCTIONS:

- Clear the offending agents and dead tissue.
- Provide growth factors for the proliferation of various cells.
- Secret cytokines that stimulate fibroblast proliferation and connective tissue synthesis and deposition.



CUTANEOUS WOUND HEALING

HEALING BY PRIMARY UNION OR BY FIRST INTENTION

DEFINITION:

- Healing of a **clean, uninfected surgical incision** in the skin joined with surgical sutures is known as **healing by primary union or by first intention**.
- The disruption of epithelial basement membrane continuity is also minimal.
- Relatively thin scar.

STAGES:

FIRST 24 HOURS:

FORMATION OF BLOOD CLOT:

- contains trapped red cells but also fibrin, fibronectin and complement components.
- acts as a scaffold for migrating and proliferating cells.
- Dehydration at the external surface of the clot leads to formation of a scab over the wound.

NEUTROPHIL INFILTRATION:

- Within 24 hours of wound, neutrophils appear at the margins.

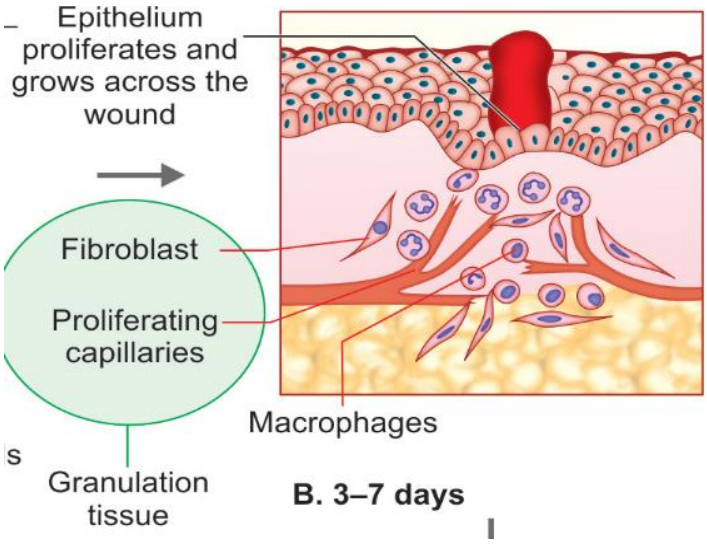
- They release proteolytic enzymes which clean out debris.

EPITHELIAL CHANGES:

- Epithelial cells from both the edges of wound proliferate and migrate towards the incisional space in the form of **epithelial spurs**.

TWO DAYS:

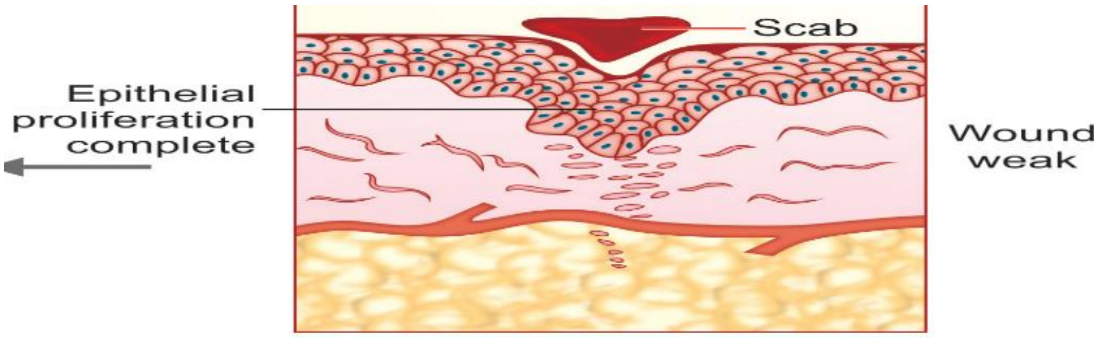
- Neutrophils are replaced by macrophages.
- Epithelial cells fuse in the midline below the surface scab
- Epithelial continuity is re-established in the form of a thin continuous surface layer.



THREE TO SEVEN DAYS:

- Progressively grows into the incision space/wound and fills the wound area by 5–7 days.
- Collagen is progressively laid down.
- Achieves its normal thickness and differentiation.
- It matures with surface keratinization.
- Inflammatory response begins to subside.

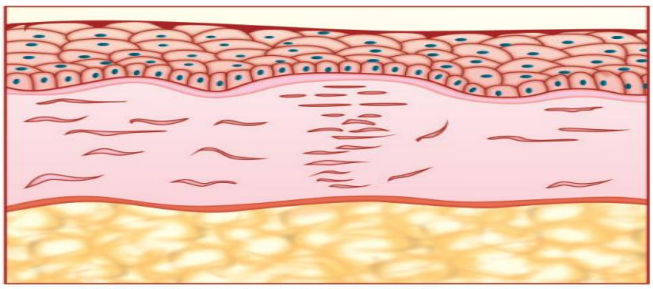
TEN TO FOURTEEN:



- Increased accumulation of collagen and regression of vascular channels.
- Granulation tissue scaffolding is converted into a pale, avascular scar.
- Wound normally gains about 10% strength of normal skin. Fibroblast proliferation occurs with collagen deposition.

WEEKS TO MONTHS:

- The scar appears as a cellular connective tissue covered by intact epidermis and without inflammatory infiltrate.
- Collagen deposition along the line of stress and would gradually achieves maximal 80% of tensile strength of normal skin.


Minimal scar with good fibrous union

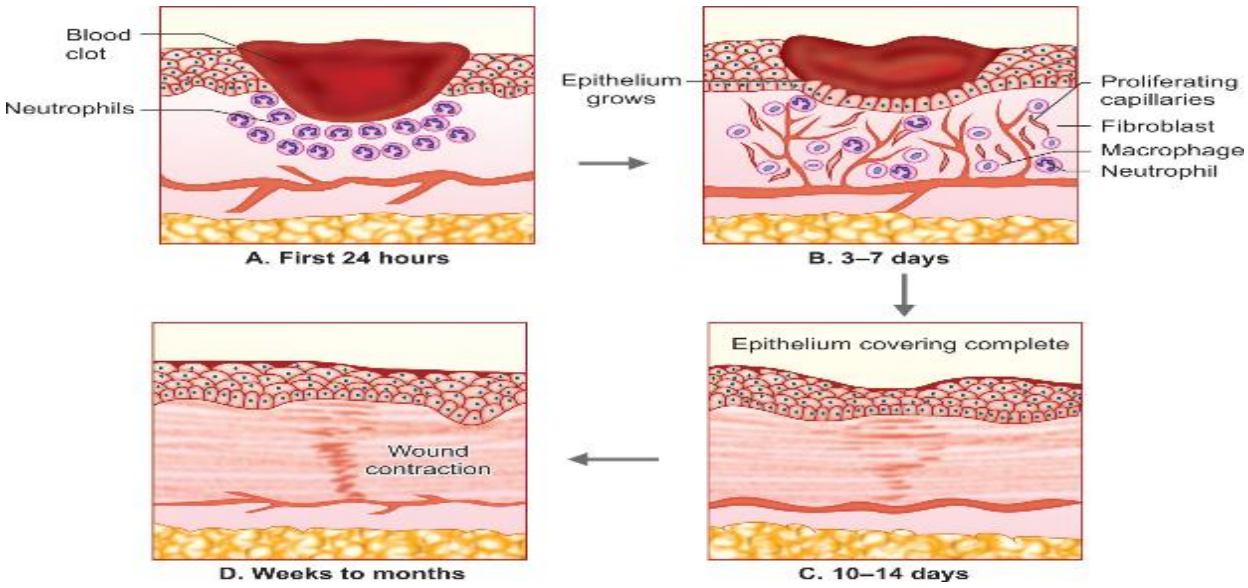
D. Weeks to months

FACTORS

PROMOTE WOUND HEALING	DELAY WOUND HEALING
• Clean wounds with closely apposed edges (sutured wound) • No infection • Good blood supply to the region • Good nutrition including vitamin C • Young age • No metabolic abnormality • Good circulatory status	• Infection • Mechanical factors • Foreign bodies • Large wounds • Wound over skin covering bone • Poor blood supply • Ionizing radiation • Nutritional deficiency • Old age • Metabolic diseases • Steroid use • Haematological abnormalities

HEALING BY SECONDARY UNION OR BY SECOND INTENTION

- Large defects on the skin surface with extensive loss of cells and tissue, the healing is more complicated.
- Basic mechanisms of healing by primary (first intention) and secondary (second intention) union are similar.



HEALING BY SECONDARY INTENTION:

- There is significant loss of tissue and the edges are far apart. Acute inflammation develops both at the edges and base.
- The cell proliferation starts from the edges and large amount of granulation tissue is formed.
- The wound is covered on the entire surface by the epithelium. The collagen fibers are deposited.
- Granulation tissue is replaced by a large scar. There is a significant wound contraction

FEATURES OF HEALING BY SECONDARY INTENTION:

- Larger wounds show more exudate and necrotic tissue.
- The clot or scab formed at the surface of wound is large.
- Full epithelialization of the wound surface is slow because of the larger gap.
- Severe inflammatory reaction because of larger defect and greater necrotic tissue.
- The larger defect requires more amount of (abundant) granulation tissue.
- Extensive deposition of collagen with substantial scar formation.

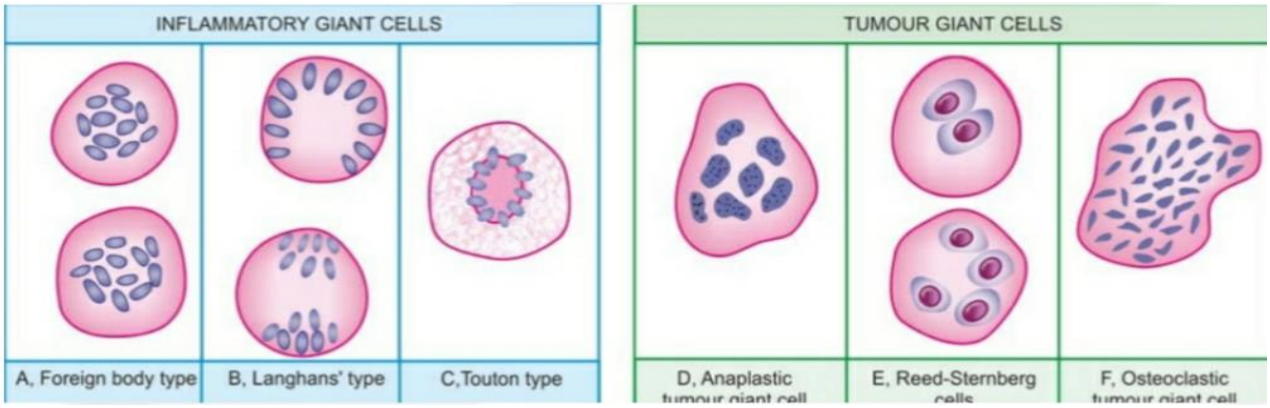
SHORT ESSAYS

1. GIANT CELLS

GENERAL FEATURES:

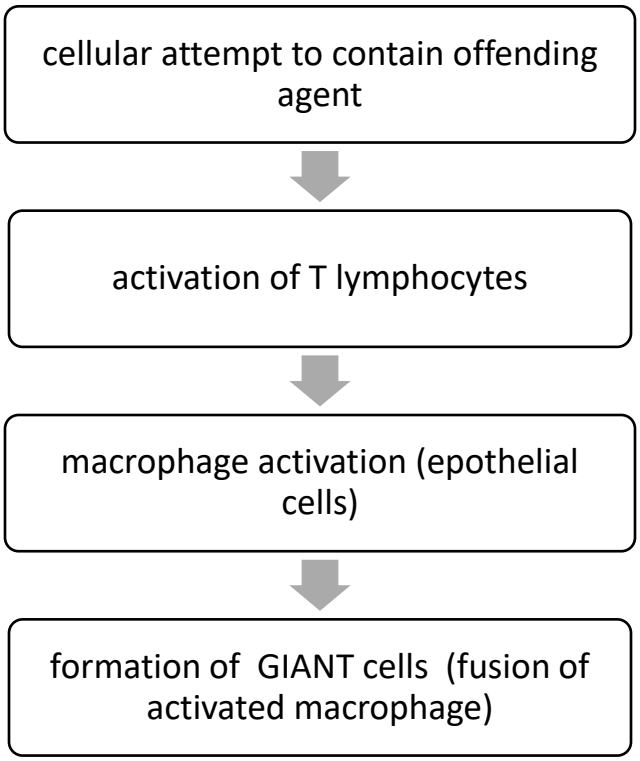
- Large in size
- Multi nucleated
- Phenotype depends on character of cell from which it was derived

Macrophage derived giant cells	Epidermal cell derived giant cells
Langhans giant cells Formed due to fusion of epithelioid cells E.g.: TB, leprosy **Foreign body giant cells** Formed due to fusion of macrophages Eg: in response to exogenous substances **Touton giant cells** Eg: xanthoma, dermatofibroma **Aschoff giant cells** Formed by fusion of 2 or more nuclei of myocardial macrophages Eg: RHD	**Tzanck giant cells** Formed due to viral infection of skin Eg: herpes simplex **Multinucleated epidermal giant cells** Eg: chronic eczema



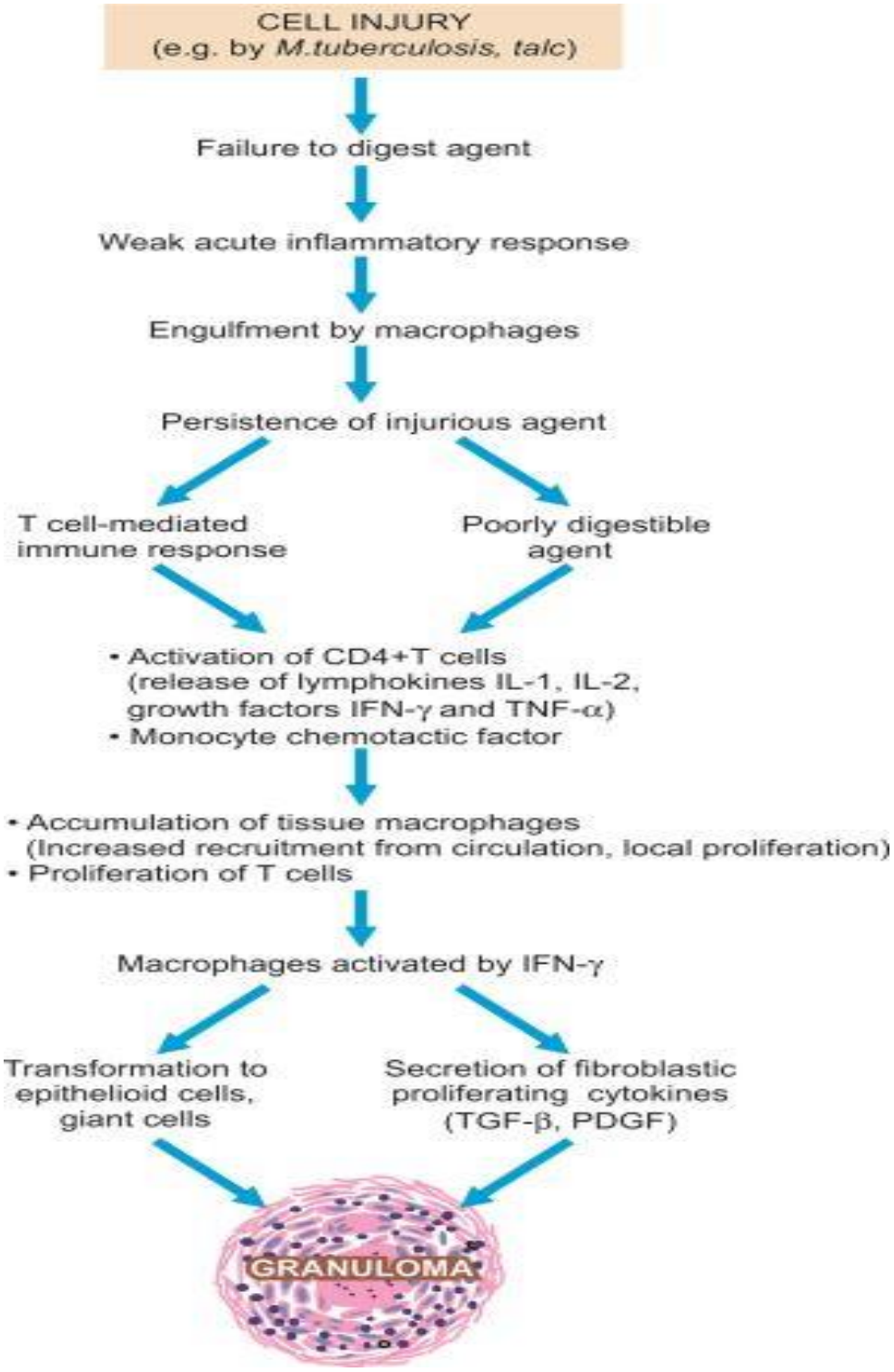
2. GRANULOMATOUS INFLAMMATION

- Pattern of chronic inflammation.
- Collection of activated macrophages (Epithelioid cells), T lymphocytes and sometimes central necrosis.



PATHOGENESIS:

- It is a type IV granulomatous hypersensitivity reaction.
- Generally, a protective defence reaction by the host but causes tissue destruction because of persistence of poorly digested antigen.


CELL INJURY
(e.g. by M.tuberculosis, talc)
Failure to digest agent
Weak acute inflammatory response
Engulfment by macrophages
Persistence of injurious agent
T cell-mediated immune response
Poorly digestible agent
• Activation of CD4+T cells (release of lymphokines IL-1, IL-2, growth factors IFN-γ and TNF-α)
• Monocyte chemotactic factor
• Accumulation of tissue macrophages (Increased recruitment from circulation, local proliferation)
• Proliferation of T cells
Macrophages activated by IFN-γ
Transformation to epithelioid cells, giant cells
Secretion of fibroblastic proliferating cytokines (TGF-β, PDGF)
GRANULOMA

TYPES OF GRANULOMA:

FOREIGN BODY GRANULOMA:

- Incited by inert foreign bodies
- Absence of T cell mediated response
- No specific inflammatory or immune response; phagocytosis occurs

IMMUNE GRANULOMA:

- Agents capable of inducing immune response (persistent microbe or self-antigen)
- T cell mediated immune response
- Macrophages activated, epithelioid cells and Giant cells (fusion of activated macrophages) formed

MORPHOLOGY:

- Epithelioid cells
- Collar of lymphocytes
- Fibroblasts and connective tissues (in older granulomas)
- Langhans giant cells
- Caseous necrosis

EXAMPLES: -

DISEASE WITH GRANULOMATOUS INFLAMMATION:

- Tuberculosis – caseating granuloma with a classical central focal necrosis
- Leprosy – non-caseating granuloma
- Syphilis - Gumma
- Sarcoidosis – non-caseating
- Cat – scratch disease – Rounded or stellate caseating
- Crohn disease – non-caseating

3. LEUKOCYTES ENDOTHELIAL ADHESION MOLECULE:

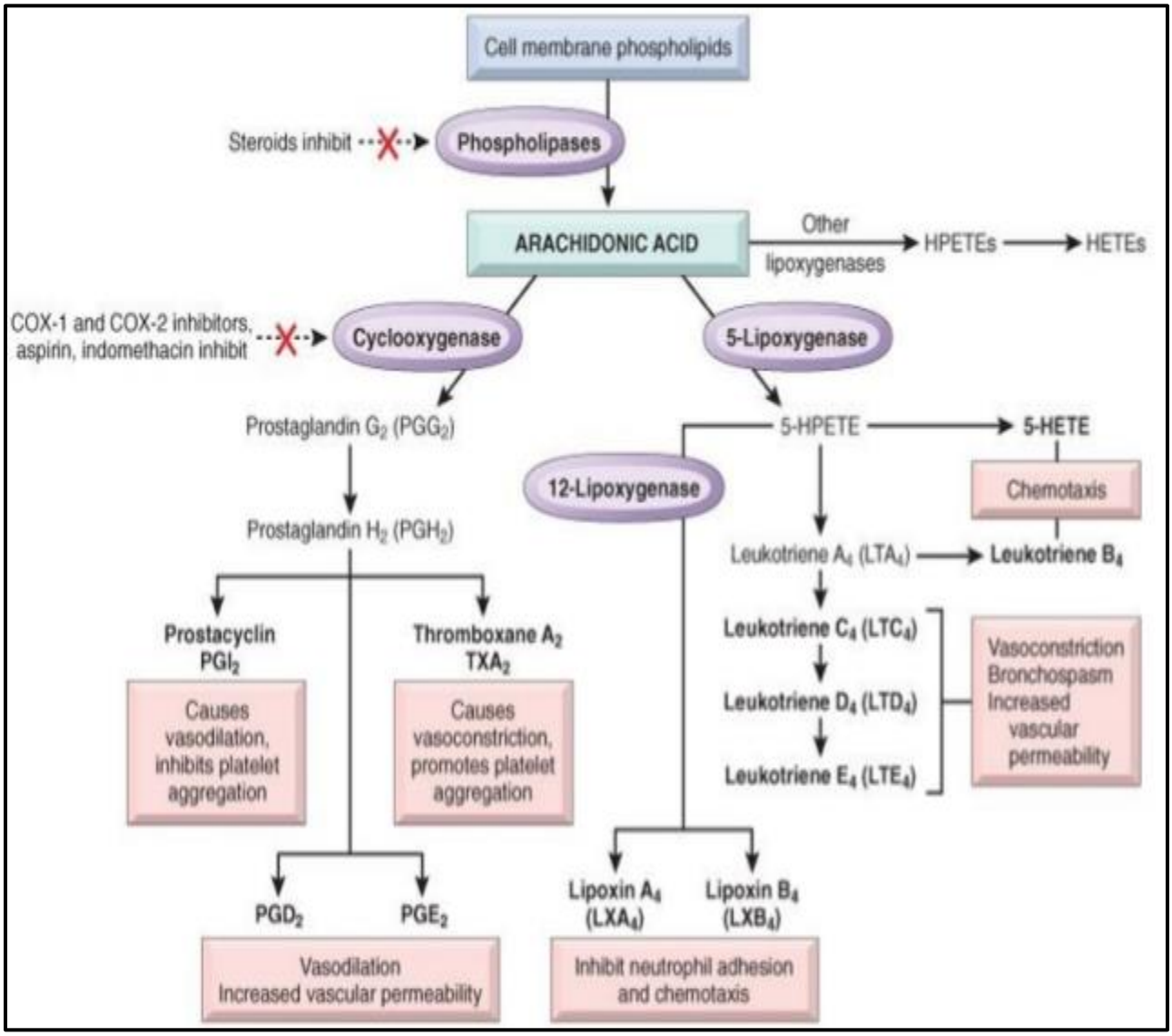
The major adhesion molecules are listed below.

- Selectins (E, P, and L) - bind via lectin (sugar-binding) domains to oligosaccharide on cell surface glycoproteins. It mediates rolling.
- Immunoglobulin family molecule (ICAM-1 and VCAM -1): these bind integrins on leucocytes and mediate adhesion.
- The principal integrins that bind ICAM-1 are beta2 integrins LFA-1 and Mac-1; the principal integrins that binds to VCAM-1 is the Beta1 integrin VLA4.

Endothelial Molecule	Leukocyte Molecule	Major Role
P-selectin	Sialyl-Lewis X–modified proteins	Rolling (neutrophils, monocytes, T lymphocytes)
E-selectin	Sialyl-Lewis X–modified proteins	Rolling and adhesion (neutrophils, monocytes, T lymphocytes)
GlyCam-1, CD34	L-selectin*	Rolling (neutrophils, monocytes)
ICAM-1 (immunoglobulin family)	CD11/CD18 (β_2) integrins (LFA-1, Mac-1)	Adhesion, arrest, transmigration (neutrophils, monocytes, lymphocytes)
VCAM-1 (immunoglobulin family)	VLA-4 (β_1) integrin	Adhesion (eosinophils, monocytes, lymphocytes)
CD31 (PECAM)	CD31	Leukocyte migration through endothelium

4. ROLE OF ARACHIDONIC ACID METABOLITE IN INFLAMMATION:

- Arachidonic acid metabolites of eicosanoids are the most potent mediators of inflammation and are derived from arachidonic acid released from cell membranes via cyclooxygenase and lipoxygenase pathway.



5. DEFECTS IN GLUCOSIDE FUNCTION:

Inherited defects in leukocyte adhesion –

- Individuals with leukocyte adhesion deficiency type 1 have a defect in the biosynthesis of the Beta-2 chain shared by the LFA-1 and Mac-1 integrins.
- Leukocyte adhesion deficiency type 2 is caused by the absence sialyl-Lewis X, the fucose-containing ligand for E and P-selectins, as a result of a defect in a fucosyl transferase, the enzyme that attaches fucose moieties to protein backbones.
- The major clinical problem in both conditions is recurrent bacterial infections due to inadequate granulocyte function.

DISEASE	DEFECTS
Leukocyte adhesion deficiency 1	Defective leukocyte adhesion because of beta chain of CD11/CD18 integrins.
Leukocyte adhesion deficiency 2	Defective leukocyte adhesion because of mutations in fucosyl transferase required for synthesis of sialyted oligosaccharide (receptor for selectins).

6. LEUKOTRIENE:

- Leukotrienes are a family of eicosanoid inflammatory mediators produced in leukocytes by the oxidation of arachidonic acid (AA) and the essential fatty acid eicosapentaenoic acid (EPA) by the enzyme arachidonate 5-lipoxygenase.

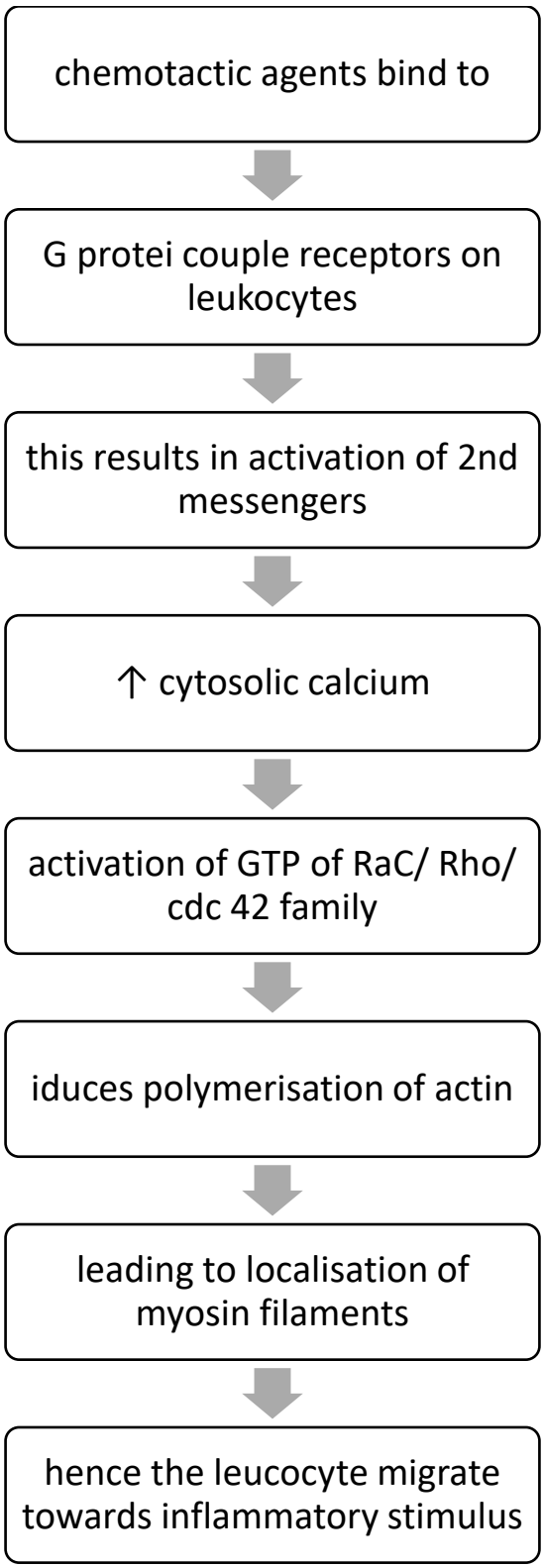
- Leukotrienes use lipid signalling to convey information to either the cell producing them (autocrine signalling) or neighboring cells (paracrine signalling) in order to regulate immune responses.
- The production of leukotrienes is usually accompanied by the production of histamine and prostaglandins, which also act as inflammatory mediators
- One of their roles (specifically, leukotriene D4) is to trigger contractions in the smooth muscles lining the bronchioles; their overproduction is a major cause of inflammation in asthma and allergic rhinitis.
- LTB_4 is a potent chemotactic agent for neutrophils.
- Vasoconstriction is brought by LTC_4, LTD_4, LTE_4.
- Leukotriene antagonists are used to treat these disorders by inhibiting the production or activity of leukotrienes.

7. CHEMOTAXIS:

- Chemotaxis (from chemo- + taxis) is the movement of an organism in response to a chemical stimulus.
- After exiting the circulation, leucocytes move in tissues to the site of injury by a process called chemotaxis

It can be both

- Exogenous agent – bacterial products
- Endogenous agent – cytokines, complementary system components, etc.


chemotactic agents bind to
G protei couple receptors on leukocytes
this results in activation of 2nd messengers
↑ cytosolic calcium
activation of GTP of RaC/ Rho/ cdc 42 family
iduces polymerisation of actin
leading to localisation of myosin filaments
hence the leucocyte migrate towards inflammatory stimulus

VERY SHORT ANSWERS

1. LIPOXINS

- Generated from arachidonic acid by lipoxygenase pathway
- Supress inflammation by inhibiting the recruitment of leukocytes Inhibit neutrophil chemotaxis and adhesion to endothelium

2. CELL DERIVED MEDIATORS OF INFLAMMATION

- Vasoactive amines – serotonin, histamine
- Arachidonic acid metabolites
- Lysosomal components
- Platelet activating factor
- Cytokines
- Nitric oxide and oxygen metabolites

3. ROLE OF VITAMIN – C IN WOUND HEALING

Vitamin C is essential for COLLAGEN SYNTHESIS in healing. In deficiency of vitamin c there is decrease in cross linking of tropocollagen. Its deficiency also decreases tensile strength of wound.

4. WOUND CONTRACTION

- Wound contraction is an important feature of healing by secondary intention and is mediated by myofibroblasts.

- Myofibroblast of granulation tissue have ultrastructural features of smooth muscle cells.

ADVANTAGES OF WOUND CONTRACTION:

- Decreases the gap between its dermal edges of the wound
- Reducing the wound surface area and thus rapid healing

5. WOUND STRENGTH:

- Fibrillar collagens (mostly type I collagen) are responsible for the development of strength in healing wounds.
- Strength results from the **increased collagen synthesis excessing degradation** and its **structural modification** (cross linking, increased fibre size).

WOUND STRENGTH:

- 10% after 1st week
- Rapidly increases during next 4 week
- 70% at the end of 3rd month.

6. ACUTE PHASE REACTANTS

- Plasma proteins which increase several hundred folds as a part of inflammatory stimulus
- Eg: CRP, fibrinogen, etc.

7. SYSTEMIC EFFECTS OF INFLAMMATION

→ Fever
→ Acute phase proteins
→ Leucocytosis
→ Septic shock
→ Lymphadenitis

8. KELOIDS

If scar tissue grows beyond the boundaries of the original wound and does not regress, it is known as keloids

9. EXUDATE VS TRANSUDATE

Exudate	Transudate
Inflammatory edema	Non inflammatory edema
Protein content is low	Protein content is high
Consists of many inflammatory cells	Consists of few cells and cellular debris

UPDATES FROM ROBBINS: 10th EDITION

1. Properties of neutrophils and macrophages:

PROPERTY	NEUTROPHIL	MACROPHAGE
ORIGIN	Hematopoietic Stem Cells (HSC) in bone marrow	• HSCs in bone marrow (in inflammatory reactions) • Many tissue-resident macrophages: stem cells in yolk sac or fetal liver (early in development)
LIFESPAN IN TISSUES	Several days	• Inflammatory macrophages: days or weeks • Tissue-resident macrophages: years
RESPONSES TO ACTIVATING STIMULI	• Rapid • Short-lived • Mostly degranulation and enzymatic activity	• More prolonged • Slower • Often dependent on new gene transcription
REACTIVE OXYGEN SPECIES	Rapidly induced by assembly of phagocyte oxidase (respiratory burst)	Less prominent
NITRIC OXIDE	Low levels or none	Induced following transcriptional activation of iNOS
DEGRANULATION	Major response; induced by cytoskeletal rearrangement	Not prominent
CYTOKINE PRODUCTION	Low levels or none	Major functional activity; requires transcriptional activation of cytokine genes
NET FORMATION	Rapidly induced, by extrusion of nuclear contents	No
SECRETION OF LYSOSOMAL ENZYMES	Prominent	Less

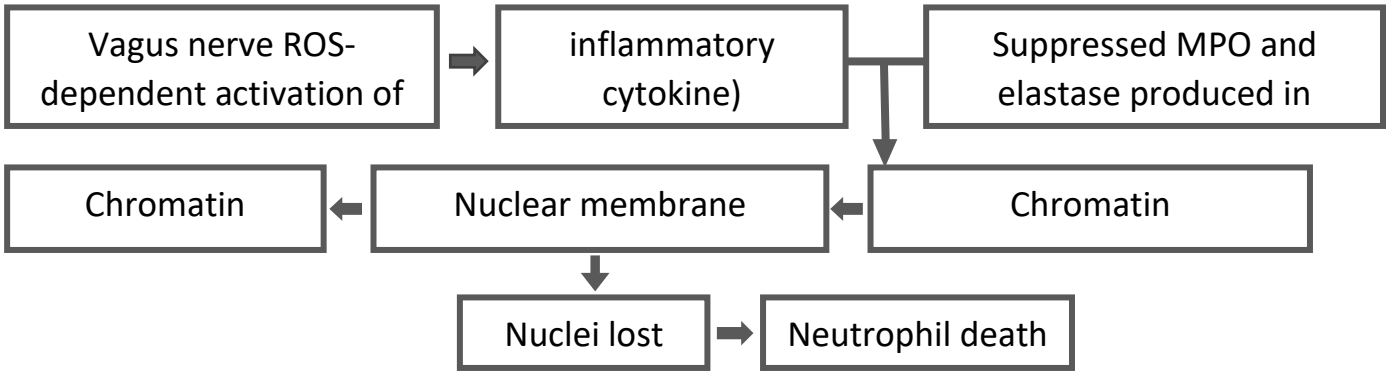
2. Names of chemokines:

- **CXCL 8** : IL-8
- **CCL 2** : Monocyte Chemoattractant Protein (MCP-1)
- **CCL 11** : Eotaxin
- **CCL 3** : Macrophage Inflammatory Protein - 1α (MIP-1α)

- **XCL1** : Lymphotactin
- **CX3CL 1** : Fractalkine

3. Formation of NET:

- Neutrophil extracellular traps (NETs) are extracellular fibrillar networks that concentrate antimicrobial substances at sites of infection and trap microbes, helping to prevent their spread.



4. Complement Factor H:

- It is a circulating glycoprotein
- It inhibits the alternative pathway of complement activation by promoting the cleavage and destruction of C3b and the turnover of the C3 convertases.
- Polymorphisms in the Factor H gene is linked to age-related macular degeneration, an important cause of vision loss in older adults.
- **Atypical Hemolytic Uremic Syndrome:**
  - It occurs due to inherited defects in Factor H and regulatory proteins that interact with Factor H (membrane cofactor protein – CD46; Factor I)
  - Here, complement deposits in glomerular vessels, leading to endothelial damage and formation of platelet-rich thrombi.

5. Role of neuropeptides in inflammation:

- Neuropeptides are secreted by sensory nerves and leukocytes
- They have a role in the initiation and regulation of inflammatory responses.
- Eg: Substance P, Neurokinin A
- Leukocytes express receptors for many neuropeptides
- The neuropeptides interact with these receptors thereby forming a mechanism for cross-talk between the nervous system and immune and inflammatory reactions



- The above observation has led to clinical trials of vagus nerve stimulation in patients with rheumatoid arthritis.

6. Fibrinopeptides:

- These are the cleavage products of fibrin
- It can stimulate inflammation
- It forms an evidence that products of coagulation mediate inflammation

7. Fig: Macrophage-Lymphocyte Interaction in Chronic Inflammation:

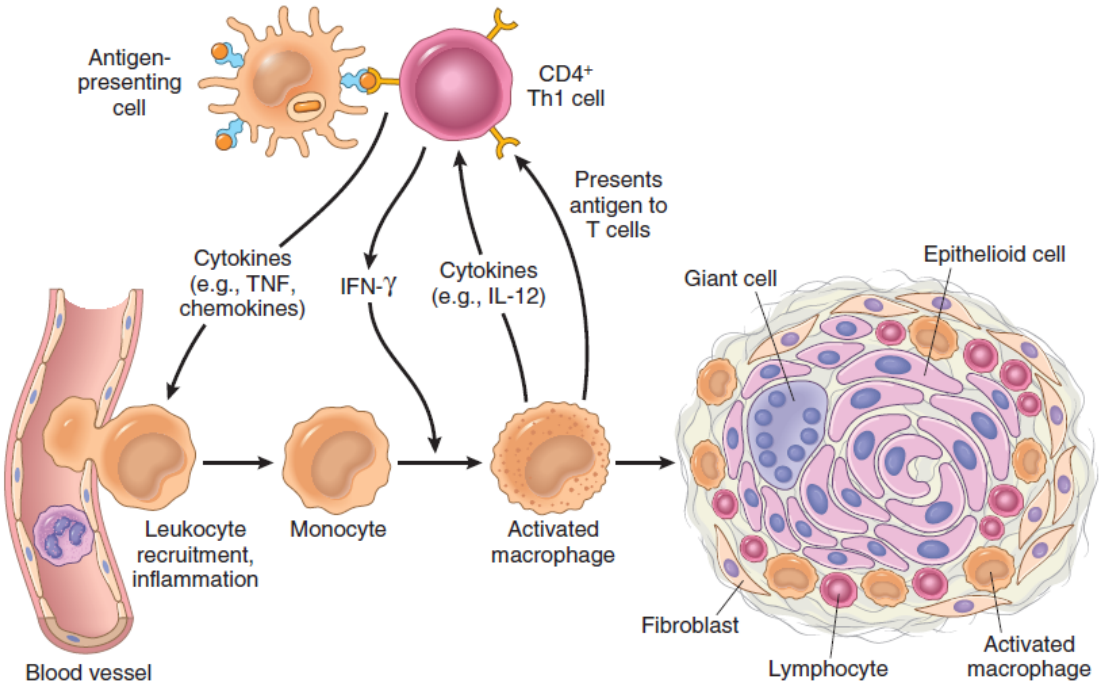


Figure 3.21 Macrophage–lymphocyte interactions in chronic inflammation. Activated T cells produce cytokines that recruit macrophages (tumor necrosis factor *[TNF]*, interleukin-17 *[IL-17]*, chemokines) and others that activate macrophages (interferon-γ *[IFN-γ]*). Activated macrophages in turn stimulate T cells by presenting antigens and via cytokines such as IL-12. Prolonged reactions involving T cells and macrophages may result in granuloma formation.

8. Role of hypothalamus in fever:

- Exogenous pyrogens act by stimulating the immune cells to release cytokines like IL-1 and TNF
- These cytokines are sensed by vascular and perivascular cells of hypothalamus
- PGE2 (Prostaglandins) released by these cells stimulate the production of neurotransmitters by the hypothalamus
- These neurotransmitters reset the body's steady-state temperature to a higher level

by

- Reducing heat loss – via vasoconstriction
- Increasing heat generation – through effects on brown fat and skeletal muscle

9. Steps is scar formation: (a major part changed)

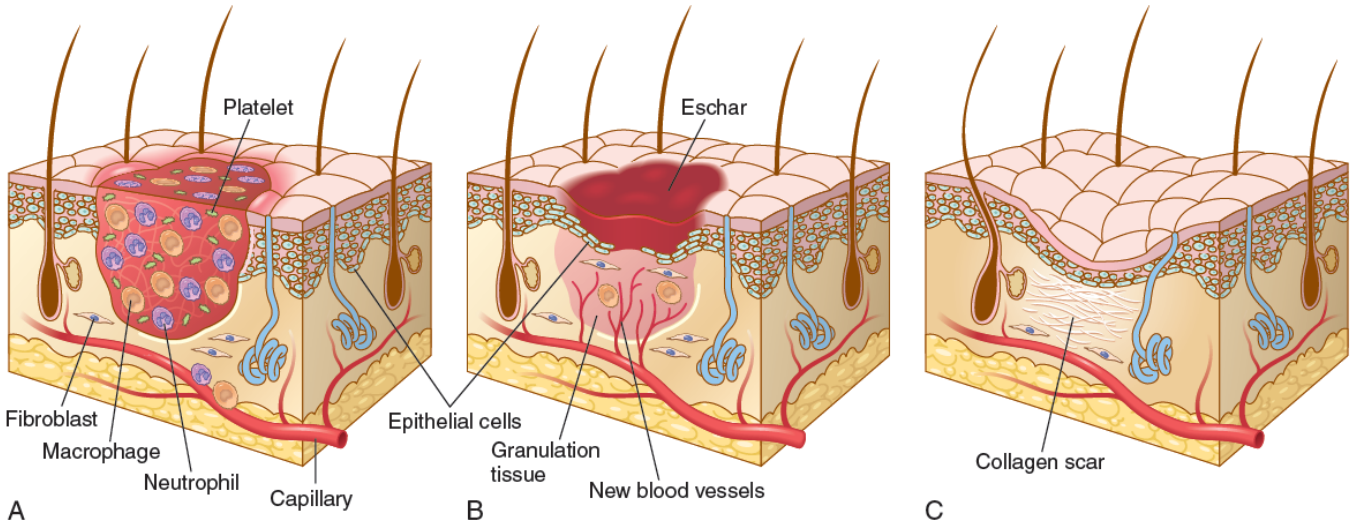


Figure 3.26 Steps in repair by scar formation: wound healing in the skin. (A) Inflammation. (B) Proliferation of epithelial cells; formation of granulation tissue by vessel growth and proliferating fibroblasts. (C) Remodeling to produce the fibrous scar.

Repair by connective tissue deposition consists of sequential processes that follow tissue injury. Within minutes after injury, a hemostatic plug composed of platelets is formed, which stops bleeding and provides a scaffold for the deposition of fibrin.

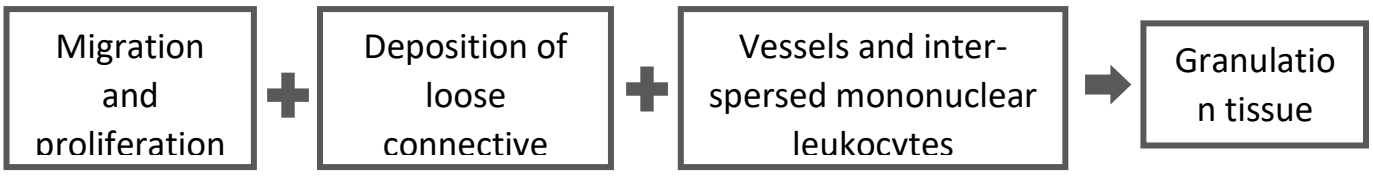
➢ Inflammation: (6 to 48 hours)

- Chemotactic agents:
  - Breakdown products of complement activation
  - Chemokines released from activated platelets
  - Other mediators produced at the site of injury
- These agents to recruit neutrophils and then monocytes which:
  - Eliminate the offending agents (microbes) that entered the wound
  - Clear the debris
- As the injurious agents and necrotic cells are cleared, the inflammation resolves.

➢ Cell proliferation: (10 days)

- After inflammation subsides, several cell types proliferate and migrate to close the now-clean wound. Each cell type serves unique functions.
- Epithelial cells respond to locally produced growth factors and migrate over the wound to cover it up.
- Endothelial cells and pericytes proliferate to form new blood vessels, a process known as angiogenesis.
- Fibroblasts proliferate and migrate into the site of injury and lay down collagen fibers that form the scar.

➢ Formation of granulation tissue:



- The term granulation tissue derives from the pink, soft, granular gross appearance.
- Histologic appearance:
  - Proliferation of fibroblasts and
  - New thin-walled, delicate capillaries (angiogenesis) in a loose ECM
  - With admixed inflammatory cells, mainly macrophages.
- Granulation tissue progressively fills the site of injury
- The amount of granulation tissue that is formed depends on:
  - Size of the tissue defect created by the wound
  - Intensity of inflammation.

➢ Deposition of connective tissue:

- Granulation tissue is progressively replaced by deposition of collagen.
- The amount of connective tissue increases in the granulation tissue, resulting in the formation of a stable fibrous scar.

10. Thrombopoietin as an Acute Phase Protein:

- Thrombopoietin is a major growth factor for megakaryocytes (platelet precursors)
- Hepcidin and Thrombopoietin (liver proteins) are released in increased amounts as part of the acute phase response, that often lead to their altered blood counts
- Upregulated Thrombopoietin as a result systemic inflammation is associated with an elevated platelet count (thrombocytosis).

11. Defects in Healing: Chronic wounds

➢ Venous leg ulcer:

- It develops most often in elderly people
- Causes:
  - Severe varicose veins
  - Congestive Heart Failure

  } Chronic Venous Hypertension
- Red cell breakdown → Deposition of hemosiderin (iron pigment) – common.
- Accompanying chronic inflammation is also seen.
- They fail to heal because of poor delivery of oxygen to the site of ulcer.

➢ Arterial ulcers

- It develops in individuals with atherosclerosis of peripheral arteries, especially associated with diabetes.
- The ischemia results in atrophy and then necrosis of the skin and underlying tissues.
- These lesions are quite painful.

➢ Diabetic ulcer

- It affects the lower extremities, particularly the feet
- There is tissue necrosis and failure to heal as a result of vascular disease.
- It results in
  - Ischemia
  - Neuropathy
  - Systemic metabolic abnormalities
  - Secondary infections.
- Histologically, these lesions are characterized by:
  - Epithelial ulceration
  - Extensive granulation tissue in the underlying dermis

➢ Pressure sores

- These are areas of skin ulceration and necrosis of underlying tissues.
- It is caused by prolonged compression of tissues against a bone,
- Eg: in elderly patients with numerous morbidities lying in bed without moving.
- They are caused by mechanical pressure and local ischemia.

CLICK HERE FOR FEEDBACK